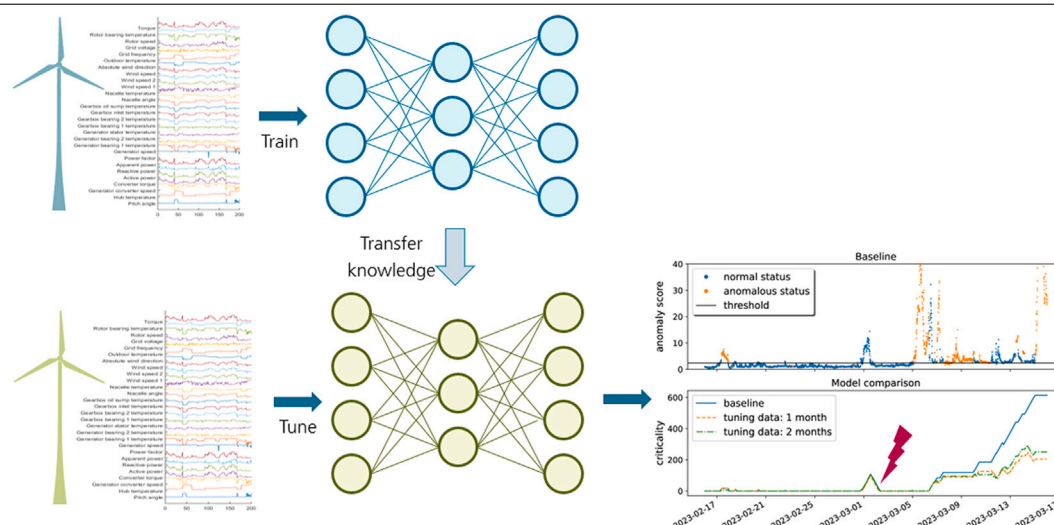


Transfer learning applications for autoencoder-based anomaly detection in wind turbines

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GRAPHICAL ABSTRACT



HIGHLIGHTS

- Transfer learning methods for anomaly detection in wind turbines are studied.
- Comparison of transferred models tuned on small data to baseline based on large data.
- Transferred models can detect anomalies leading to faults as well as the baseline.
- Fine-tuning of decoder parameters improves the performance without overfitting.
- Transferred models are able to detect anomalies before failures in real-world data.

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ABSTRACT

Anomaly detection in wind turbines typically involves using normal behaviour models to detect faults early. Normal behaviour models are often implemented through the use of neural networks, of which autoencoders are particularly popular in this field. However, training autoencoder models for each turbine is time-consuming and resource intensive. Thus, transfer learning becomes essential for wind turbines with limited data or applications with limited computational resources. This study examines how cross-turbine transfer learning can

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Fault detection

be applied to autoencoder-based anomaly detection. Here, autoencoders are combined with constant thresholds for the reconstruction error to determine if input data contains an anomaly. The models are initially trained on one year's worth of data from one or more source wind turbines. They are then fine-tuned using small amounts of data from the target wind turbine. Three methods for fine-tuning are investigated: adjusting the entire autoencoder, only the decoder, or only the threshold of the model. The performance of the transfer learning models is compared to baseline models that were trained on one year's worth of data from the target wind turbine. The results of the tests conducted in this study indicate that models trained on data of multiple wind turbines do not improve the anomaly detection capability compared to models trained on data of one source wind turbine. In addition, modifying the model's threshold can lead to comparable or even superior performance compared to the baseline, whereas fine-tuning the decoder or autoencoder further enhances the models' performance.

Nomenclature

WT	Wind turbine
AE	Autoencoder
NBM	Normal behaviour model
SCADA	Supervisory Control and Data Acquisition
TL	Transfer Learning
OP-mode	Operational mode
RMSE	Root mean square error

1. Introduction

Wind power plays a crucial role in addressing climate change, offering significant promise among renewable energy sources. However, wind farms are often situated in regions with challenging weather conditions, leading to complex operating conditions and potential failures in wind turbines (WTs). This results in increased maintenance costs and decreased efficiency. To mitigate these challenges, there has been a growing focus on condition monitoring and early fault detection using machine learning models for WTs [1]. Numerous approaches concentrate on training a normal behaviour model (NBM) using WT Supervisory Control and Data Acquisition (SCADA) data, to identify anomalous behaviour or detect specific failures in WTs. Examples of anomaly detection in WT SCADA data can be found in [2–17]. Often, AE-models are used as the core of the NBM [4–7,10,13–17] and have shown promising results when applied to high dimensional SCADA data.

Training an anomaly detection model based on a NBM such as an AE model requires enough data representing normal operation of the WT. However, such data is not available for new installations or after maintenance actions, such as an oil refill or component replacements resulting in changed normal behaviour. Also, anomaly detection in WTs usually involves training a NBM for each WT separately, which is time-consuming and resource-intensive. To overcome both of these problems, Transfer Learning (TL) can be used.

TL involves transferring learned behaviour from one model to another, similar to how humans learn new tasks by applying knowledge from related tasks. In the context of WTs, this involves using a pre-trained model for anomaly detection in one WT (or a group of WTs) and repurposing it for anomaly detection in another WT. This approach saves significant time and resources that would otherwise be required for training a new model [18,19]. In addition, Simani et al. have shown that accurate NBMs for WTs can be created using only a small amount of data. However, they do not investigate whether anomaly detection is still possible with these transferred models [20].

In this study, our focus is on exploring the applications of TL for AE-based anomaly detection in WT data, to test whether anomaly detection with a small amount of data is feasible. We first explore TL within one wind farm, to test whether anomaly detection is still possible for a WT for which only a small amount of training data is available.

Specifically, a model is trained on data of one WT and transferred to another WT within the same wind farm. We examine the impact of different amounts of tuning data for different WT types, using various TL methods. Additionally, we use data of multiple WTs to train a NBM and fine-tune it on data of a single target WT which is not in the training dataset. The hypothesis is that this model can generalise better and be better tuned to the target task. In both cases, we validate the transferred models' ability to detect anomalies by evaluating three case studies, demonstrating that early fault detection in WTs is indeed possible even with a limited amount of training data.

The paper is structured as follows. First we present an overview of related work in Section 2. In Section 3 we describe the data and models used and the TL methods which we applied. Next, the results are presented and discussed in Section 4. The results of the transferred models are first compared to their corresponding baseline models, to show they are capable of distinguishing normal from anomalous behaviour almost or as well as the baseline model. Afterwards, we demonstrate the model ability to detect failures early using three case studies. Finally, the paper is summarised in Section 5.

2. Related work

Research on TL applied to unsupervised or semi-supervised (only using 'healthy' data) machine learning, such as anomaly detection, is limited [21,22]. Most work focuses on dimensionality reduction and clustering, which is often not applicable to anomaly detection in predictive maintenance settings. Due to limited occurrences of failures, anomaly detection models are often hard to train. Here, TL approaches may help overcome this problem [23]. In addition, the training of WT-specific models is not always feasible. For example, if a WT is newly installed or not all historical data is informative due to re-calibration or component replacement(s). Also, training ML models on only one turbine greatly reduces the amount of training time for the whole wind farm (containing up to hundreds of turbines) [22].

Zraggen et al. present three TL approaches for anomaly detection in WTs using deep learning methods [22]. The anomaly detection model used is a normal behaviour model (NBM) of a regression CNN that has 4 SCADA parameters as input and predicts the gearbox bearing temperature. Three TL methods are tested for two scenarios: cross-turbine TL within the same wind farm and cross-turbine TL across different wind farms. For the first scenario the simplest method, where an additional linear regression (LR) is added to the pre-trained CNN, works almost as well as CNN trained from scratch on the target turbine. For the cross-wind-farm case, there are more non-linear domain shifts to consider as well. Here, they added both the LR model as well as a CNN model that predicts the error between the input and the output of the LR. Simply fine-tuning the original CNN after pretraining did not yield better results, as it can easily overfit.

Another application of TL for anomaly detection in WTs is presented by Dimitrov et al. [24]. The authors use TL to create a physics-informed machine-learning model, by transferring knowledge from a model trained on simulations to a target model trained on real data. The two models tested are a five-layer dense neural network (ANN) and a five-layer denoising autoencoder (DAE). Both use six SCADA

parameters, of which the electrical power is used as a health indicator. The prediction performance of the ANN increases significantly with TL, even when only one month of SCADA data is used for fine-tuning. The power prediction of the DAE did not improve a lot, as it already had very high performance. The anomaly detection ability of the DAE was however not tested.

To enable the use of structurally incomplete data (where only some sensor values are available due to different sample rates), Chen et al. use TL to train fault classification models for different types of incomplete data and a model based on only the complete data [23]. The information is then transferred between these models to optimally use all available data, instead of only a subset (the structurally complete data). The proposed method has a higher accuracy than the model solely trained on structurally complete data and the resulting models can be applied to all data.

Wen et al. use TL to transfer information from univariate time series models to multivariate time series models for anomaly detection [25]. The model is based on an encoder-decoder network (U-Net) and extended for multivariate time series. The model for multivariate time series has encoders for each input feature, to which the information from the models for univariate time series is transferred. The resulting TLmodels all outperformed the models without TL.

Recently, Simani et al. study TL for anomaly detection in SCADA data of WTs [20]. They focus improving the accuracy of the NBM's output signals and show that this indeed is possible between different WTs in the same wind farm. However, the application to fault diagnosis or anomaly detection is not investigated.

3. Method

In this section, the data, model and transfer learning methods are described.

3.1. Data

This study utilises data from two offshore wind farms, WF A and WF B, located in the German North Sea. These wind farms each contain WTs of a different type, i.e. the data used is different for both wind farms. The dataset used in this study is derived from SCADA data and spans several years. It consists of ten-minute-average values from 314 sensors for WF A and 58 sensors for WF B. The sensors capture various measurements, including power production, wind speed and direction, various currents and voltages, as well as temperature measurements of bearings.

In addition to the SCADA data, the dataset includes information about the operational mode (OP-mode) for each timestamp. Examples of OP-modes include normal operation, derated operation, service, downtime, and others. For the purposes of this study, only 'normal operation' is considered as normal behaviour, while all other OP-modes are categorised as 'anomalous'.

For the experiments conducted in this paper, we used data of 10 WTs from WF A and 6 from WF B. The selected WTs are spread across the wind farms, such that the dataset contains WTs on both sides as well as in the middle of the wind farm.

3.2. Anomaly detection model

For the anomaly detection model, we use a similar setup as presented in [10,16].

First, data representing normal behaviour is selected based on the OP-mode. In addition, only data with power $P > 0$. Due to communication issues the OP-mode is not always properly closed, resulting in some downtimes being missed and logged as, for example, 'normal operation'. Furthermore, we filter out any values where the power is high but the wind speed is outside the cut-in and cut-off values of the WT. The data filtered out is labelled as 'anomalous'.

Table 1

Hyperparameters of the AE models for each wind farm.

	WF A	WF B
# units in hidden layers	200, 100, 50, 100, 200	25, 10, 25
Learning rate	0.001	0.001
Epochs	10	10
Activation	PReLU	PReLU
Batch size	128	64

Next, data is prepared for the AE model. In order to enable the detection of as many anomalies in as many different parts of the WT as possible we keep as many features as possible. However, some features are found to have a negative impact on the training of an AE. Therefore, we remove any features that represent counter values such as the number of cable windings or the energy production since commissioning of the WT. Then, we eliminate set points and features that are constant or have up to 3 unique values which indicates low variance. The data as provided by the operators does not contain any missing values, although some features contain many implausible consecutive zero values, which are suspected to be originally missing values. Consequently, we also remove features that have more than 80% zero values. After feature selection there are respectively 238 and 28 features for WF A and WF B turbines left. For features representing angles, we transform them into sine and cosine values to ensure their continuity. Finally, we scale the data between 0 and 1 using min-max-scaling.

The AE model used is a fully connected feed forward neural network with a small bottleneck layer with PReLU [26] as activation function. For WF A the network consists of 7 layers, where the hidden layers have 200, 100, 50, 100 and 200 units respectively. For WF B, there are 3 hidden layers, with 25, 10, and 25 units. The given hyperparameter specifications were found best fitting through a grid search on data from the given wind farms to optimise the AE reconstruction errors of a validation dataset. The network's parameters are optimised using the Adam optimisation algorithm [27] with a learning rate of 0.001 to minimise the mean square error of the reconstruction. In order to avoid overfitting the training of the AE is set to only 10 epochs which was found to be well fitting for the given datasets through testing with validation data. An overview of the model hyperparameters for the baseline models is provided in Table 1.

To determine whether the model detected an anomaly, we calculate the root mean square error (RMSE) of the reconstruction errors for each timestamp in the training data. This score is referred to as the Anomaly Score S . Alternatively, the Anomaly Score can also be defined through the Mahalanobis distance of the reconstruction error. However, this metric requires an accurate estimate of the covariance matrix of the data, which is difficult to estimate for high dimensional data. In addition, the model performed better using the RMSE as Anomaly Score for the datasets used in this study.

Finally, we set a threshold by maximising the $F_{1/2}$ -score for the training data. The F_{β} -score is the harmonic mean of the recall

$$R = \frac{TP}{TP + FN} \quad (1)$$

and the precision

$$P = \frac{TP}{TP + FP} \quad (2)$$

Thus, the F_{β} -score is given by

$$F_{\beta} = \frac{(1 + \beta^2) \cdot P \cdot R}{\beta^2 \cdot P + R}, \quad (3)$$

where TP is the number of true positives, FN the number of false negatives, FP the number of false positives and β is a parameter that allows for weighting precision or recall as more important. In this case, $\beta = 1/2$ is chosen, to ensure high precision and reduce false

positives i.e. false alarms. The anomaly labels used for optimisation of the $F_{1/2}$ -score are based on the OP-mode and power values as described before for normal data selection. Let $S_1 \leq \dots \leq S_N \in \mathbb{R}_{\geq 0}$ be an ordered series of Anomaly Scores and $l_1, \dots, l_N$ the anomaly labels to the corresponding data points. If all labels have the same value, no meaningful threshold can be calculated. In this case the threshold is set to the value S_N . In all other cases the process of finding the optimal threshold t^* is defined by algorithm 1.

Algorithm 1 Threshold Selection Algorithm

```

 $f_{max} \leftarrow 0$ 
 $i_{max} \leftarrow 0$ 
for  $i \in \{1, \dots, N\}$  do
   $t \leftarrow S_i$  ▷ threshold candidate
  for  $j \in \{1, \dots, N\}$  do
     $p_j(t) \leftarrow S_j \geq t$  ▷ prediction based on  $t$ 
  end for
   $f \leftarrow F_1\text{-Score of } l_1, \dots, l_N \text{ and } p_1(t), \dots, p_N(t)$ 
  if  $f \geq \frac{1}{2} f_{max}$  then
     $f_{max} \leftarrow f$ 
     $i_{max} \leftarrow i$ 
  end if
end for
 $t^* \leftarrow S_{i_{max}}$ 
return  $t^*$ 

```

3.3. Transfer learning

In this paper, we use two types of source models: models trained on 12 months of data of one WT (i.e. 52560 data points) and models trained on 12 months of data of multiple WTs. From this point on, we will refer to the two types of TL models as *asset-to-asset* and *multi-asset* models. These models are then transferred to the target WT using the following three tuning methods:

- **Threshold:** the AE of the source model is applied as is. Only the anomaly score threshold is updated.
- **Decoder:** In addition to updating the threshold, the decoder of the AE is fine-tuned while keeping the encoder fixed. We use a learning rate of 0.001 and train for 10 epochs.
- **AE:** In this case the complete AE network is fine-tuned using a learning rate of 0.0001 (1/10 of original learning rate to avoid overfitting) and 10 epochs. Afterwards, the threshold is updated.

The source models based on one WT are the same as the baseline models, of which the performance is compared to that of the transferred models. The source models of the *multi-asset* models are created using the same network architecture shown in Table 1. The number of epochs for these models was set to 30.

For each of the target WTs, we use 1 to 3 months of SCADA data to fine-tune the source models, to validate whether it is possible to create an accurate anomaly detection model if only a small amount of data of the target WT is available.

The training data was selected such that there was enough data representing normal behaviour to create an AE model for each of the selected WTs. For WF A, the training data for the source and baseline models spans from March 2022 until March 2023, with April 2023 as the prediction data. For WF B, the training data spans from August 2022 until August 2023, with September 2023 as the prediction dataset. The models are fine-tuned using the data from the last month(s) before the prediction data.

3.4. Evaluation

The anomaly detection capability of the model is evaluated using the $F_{1/2}$ -score (as defined in Eq. (3)) on one month of data immediately

following the training or tuning data. We use the $F_{1/2}$ -score instead of the F_1 -score to prioritise precision over recall for our model, since operators of wind farms are primarily interested in accurately detecting anomalies while minimising false alarms. We determine whether anomalies were correctly detected by using the labels ‘anomaly’ and ‘normal’ based on the OP-mode, wind, and power measurements, as explained in Section 3.2.

We compare the performance of tuned models with a baseline model trained on 12 months of data. Furthermore, we conduct three case studies to demonstrate that the transferred models are still effective in detecting anomalous behaviour due to failures.

Here, we use the criticality C measure as defined in [10] to evaluate the anomalies detected. The criticality value increases by one if an anomaly is detected during normal operation, decreases by one if no anomaly is detected during normal operation, and remains constant if the OP-mode is not normal. The criticality is computed by the algorithm 2, where $t_1, \dots, t_N$ are the timestamps of the predictions, $s_{t_i} \in \{0, 1\}$ the status information, where 1 corresponds to a normal status and 0 to an abnormal status and $p_{t_i} \in \{0, 1\}$ the prediction, where 1 represents a detected anomaly and 0 no detected anomaly for $i = 1, \dots, N$. The range of the criticality value is limited to $[0, 1000]$.

Algorithm 2 Criticality Algorithm

```

 $crit \leftarrow [0, 0, \dots, 0] \in \mathbb{N}^{N+1}$  ▷ Init.  $crit$  with 0 at timestamp  $t_0$ 
for  $i \in \{1, \dots, N\}$  do
  if  $s_{t_i} = 0$  then
    if  $p_{t_i} = 1$  then
       $crit[i] \leftarrow \min\{crit[i-1] + 1, 1000\}$ 
    else
       $crit[i] \leftarrow \max\{crit[i-1] - 1, 0\}$ 
    end if
  else
     $crit[i] \leftarrow crit[i-1]$ 
  end if
end for
 $crit \leftarrow crit[1 : N]$  ▷ Remove timestamp  $t_0$  from  $crit$ 
return  $crit$ 

```

4. Results and discussion

First, baseline models for each of the selected 10 WTs of WF A and each of the 6 WTs of WF B were created to compare to the TL models. The baseline models were trained using 1 year of SCADA data representing normal behaviour. The performance of these baseline models is shown in Fig. 1. For WF A WTs 1 and 40 are on opposite sides of the wind farm, whereas 28 and 29 are in the middle. For WF B WTs 42 and 70 are opposite sides, WTs 55 and 56 are in the middle of the wind farm.

The source models were transferred using the methods described in Section 3.3. In Sections 4.1 and 4.2 the results for the *asset-to-asset* and *multi-asset* models are described respectively, while the case studies are described and analysed in Section 4.3. Finally, we discuss possible improvements and future research in Section 4.4.

4.1. Asset to asset transfer learning

For WF A, the source models used are the same as the baseline models of WTs 1 and 40, located on opposite sides of the wind farm. The purpose was to investigate whether the location has an influence on the model’s performance on the target WTs. We will refer to these WTs as “source WTs” from now on. For WF B, the same setup was used, using source WTs 42 and 70. These source models were transferred to all other selected WTs, i.e. 18 and 10 TL models for each tuning method were trained for wind farm A and B respectively.

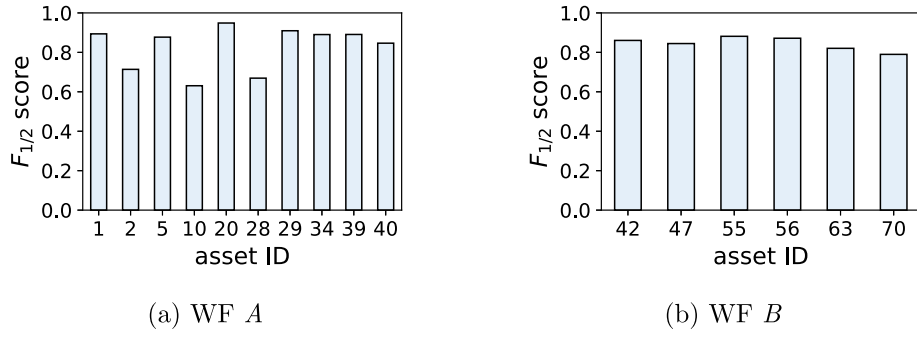
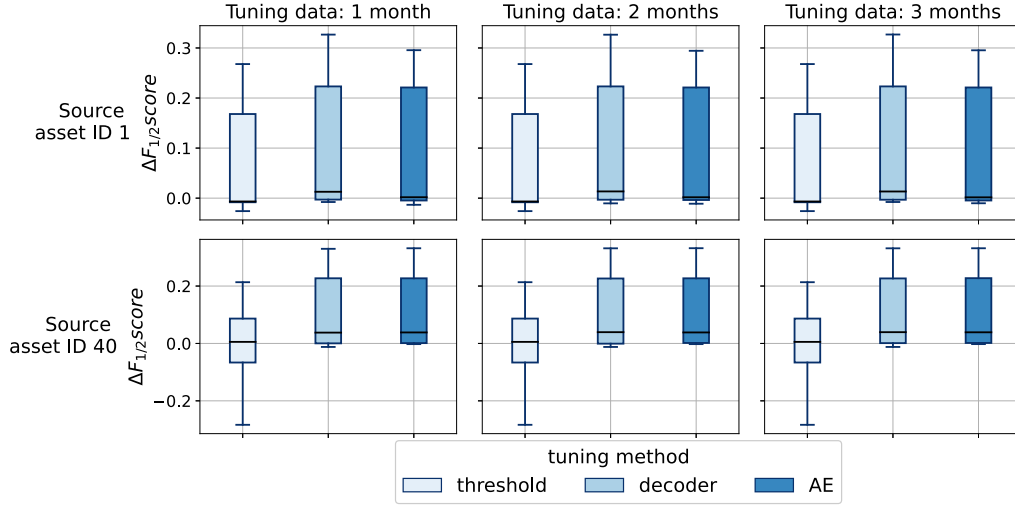
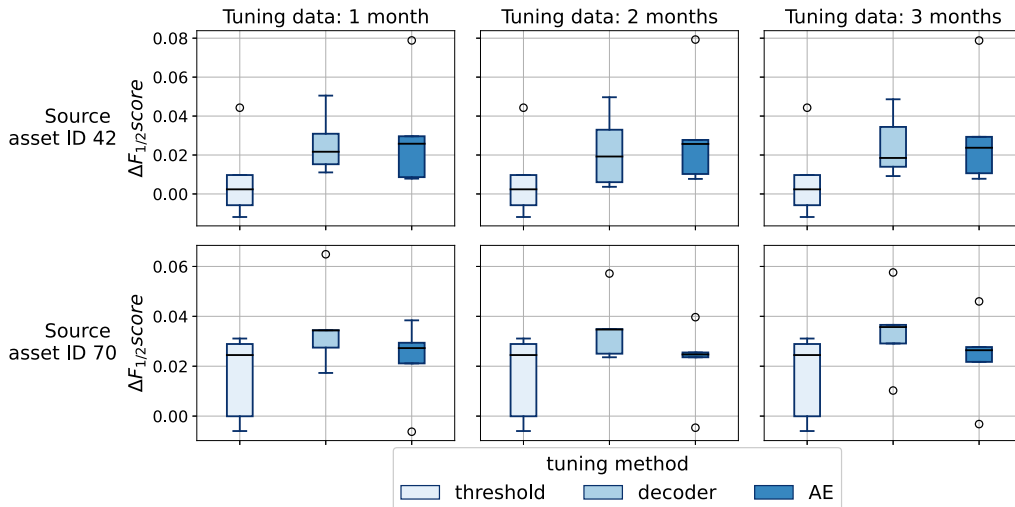


Fig. 1. Performance of the baseline models.

Fig. 2. Transfer learning results for WF A of the *asset-to-asset* models. The difference in $F_{1/2}$ score with respect to the baseline model for each wind turbine is shown. The upper row shows the increase in $F_{1/2}$ score for the transferred models w.r.t. the baseline models, using wind turbine with asset ID 1 to create a source model and fine-tuning on 1, 2 and 3 months of data of the target wind turbines respectively. The bottom row shows the same for transferred models based on source wind turbine 40.Fig. 3. Transfer learning results for WF B of the *asset-to-asset* models. The difference in $F_{1/2}$ score with respect to the baseline model for each wind turbine is shown. The upper row shows the increase in $F_{1/2}$ score for the transferred models w.r.t. the baseline models, using wind turbine with asset ID 42 to create a source model and fine-tuning on 1, 2 and 3 months of data of the target wind turbines respectively. The bottom row shows the same for transferred models based on source wind turbine 70.

Results are shown in Figs. 2 and 3 for TL models based on only a small amount of tuning data, ranging from 1 to 3 months. Notably, these results show that it is indeed possible to create an anomaly detection model with a relatively small amount of data. The $F_{1/2}$ scores of the TL models are comparable or slightly better than the baseline

models, which were trained on a full year of SCADA data. Whether 1 month or 3 months of tuning data is used, does not seem to have a significant effect. Detailed results of the *asset-to-asset* models can be found in Table 2 for WF A, where the average changes in $F_{1/2}$, precision, recall and accuracy are shown. While $F_{1/2}$, precision and

Table 2

Average change in $F_{1/2}$ -score, Precision, Recall and Accuracy for the *asset-to-asset* models w.r.t. baseline models for WF A using 3 months of tuning data.

Source	Method	$\Delta F_{1/2}$	Δ Precision	Δ Recall	Δ Accuracy
1	Threshold	0.068	0.127	-0.089	0.017
	Decoder	0.093	0.134	-0.032	0.029
	AE	0.088	0.129	-0.037	0.027
40	Threshold	0.008	0.037	-0.080	0.004
	Decoder	0.098	0.137	-0.025	0.031
	AE	0.100	0.138	-0.021	0.032

Table 3

Average change in $F_{1/2}$ -score, Precision, Recall and Accuracy for the *asset-to-asset* models w.r.t. baseline models for WF B using 3 months of tuning data.

Source	Method	$\Delta F_{1/2}$	Δ Precision	Δ Recall	Δ Accuracy
42	Threshold	0.008	0.004	0.010	0.003
	Decoder	0.025	0.015	0.040	0.011
	AE	0.030	0.033	0.020	0.010
70	Threshold	0.016	0.011	0.019	0.006
	Decoder	0.034	0.010	0.072	0.017
	AE	0.024	-0.005	0.074	0.014

accuracy all show better performance of the TL models, the recall is on average slightly lower. Thus, the *asset-to-asset* models show less false positives, but are less sensitive to anomalous behaviour than the models trained on 1 year of SCADA data of the target WT. For WF B the TL models perform slightly better than the baselines in general, as can be seen in Table 3.

Interestingly, the location of the chosen source WTs does not appear to have a significant impact on the model's performance. The TL models trained using data from WT 1 fit the data of WT 40, on the opposite side of the wind farm, quite well, even with just one month of tuning data. The same is observed for WF B. There is however not enough data to state that one can simply use any of the WTs as source, since the data quality of the source WT overshadows this effect. Therefore, we use one of the middle assets in the WFs with a good baseline model performance as source WT for the case studies.

The transferred models show promising results, performing on par or slightly better than the baseline models. It is worth noting that the AE of the source model is capable of reconstructing the data of the target WTs, even without tuning the AEs. These models can yield good model scores, even though the AE was trained on data from a different WT, showing that the WTs within the same wind farm behave very similarly, since the normal behaviour primarily depends on the environment.

In some cases, the transferred models even outperform the baseline models. The improved performance of the transferred models may be attributed to their ability to fill knowledge gaps in the baseline models, particularly when the training data of the baseline models lacks normal behaviour within a period (e.g. winter or summer) similar to the prediction period. Also, the TL models might benefit from focusing on the weeks shortly before the prediction period.

Fine-tuning the decoder or the complete AE further enhances the models' performance in terms of $F_{1/2}$ score compared to the threshold method, as depicted in Figs. 2 and 3. Specifically, fine-tuning the decoder alone outperforms the other TL methods in most cases. However, we find that caution must be exercised when tuning the AE, as it could lead to overfitting and negative transfer.

4.2. Multi-asset model transfer learning

In this section, we discuss the results for the *multi-asset* TL models. Here, the source models were trained on data from all selected WTs except for the target WT. Next, the source models were fine-tuned on a small amount of tuning data of the remaining (target) WT. For WF A 10 source models were created and 10 TL models for each tuning

Table 4

Average change in $F_{1/2}$ -score, Precision, Recall and Accuracy for the *multi-asset* models w.r.t. baseline models using 3 months of tuning data.

Wind farm	Method	$\Delta F_{1/2}$	Δ Precision	Δ Recall	Δ Accuracy
A	Threshold	-0.093	-0.094	0.003	-0.056
	Decoder	-0.084	-0.123	0.052	-0.026
	AE	-0.064	-0.101	0.043	-0.025
B	Threshold	-0.037	0.000	-0.145	-0.102
	Decoder	-0.023	0.002	-0.094	-0.065
	AE	-0.020	0.002	-0.084	-0.058

method. For WF B 6 source models were created and 6 TL models for each tuning method. Results of the *multi-asset* models are summarised in Fig. 4, while detailed metrics are shown in Table 4.

Compared to the results of the *asset-to-asset* models, these models do not perform as well, particularly for WF A. Since the WTs within one wind farm behave very similar, this could be caused by overfitting, as the model sees almost the same data multiple times. However, the learning curves do not show that this is the case, as seen in Fig. 5. Here, learning curves of a *multi-asset* source model and the fine-tuned models based on this model are shown. It is likely that the models are not sensitive enough to detect anomalies properly. Table 4 also shows that the change in precision for TL models is negative for WF A and almost zero for WF B. Also, the accuracy is reduced, which is unusual for less sensitive anomaly detection models, since anomalies are the minority class. Overall, the *multi-asset* models do not distinguish normal from anomalous behaviour well.

In case of WF A, the TL models for all WTs except 40 perform significantly better with 3 months of tuning data. For WF B the models' performance does not seem to be significantly influenced by the amount of tuning data.

4.3. Case studies

For the case studies we use the same set up as before, where a baseline model is trained on one year of data of the target WT to compare with TL models. The source models are trained with data of a WT in the middle of the wind farm, for which the baseline model performed well. For WF A this is WT 29, for WF B WT 55. Next, the models' decoders were fine-tuned with 1 to 2 months of data of the target WTs. The first two case studies are for WTs of WF A, the third case study contains a failure in a WT in WF B.

4.3.1. Case study 1

Anomalies are detected from 2022-12-22 onward, which were caused by problems in the rotor brake and hydraulic system. The anomalies are caused by several related sensor measurements suddenly behaving very differently. Both TL models are able to detect this anomaly easily, as seen in Fig. 6. The anomaly score during abnormal operation on 2022-12-27 is lower for the TL models compared the anomaly score of the baseline model, indicating that the TL models are less sensitive than the baseline model. This is however expected, since only a small amount of target data was used for tuning.

4.3.2. Case study 2

Problems in the converter cooling system causes a sensor drift, which in turn causes a positive trend in the anomaly score of the baseline model, as seen in Fig. 7. After maintenance on 2022-10-07 the problem was fixed and the anomaly score normalises. Both tuned models are able to detect the problem, although they detect the anomaly later than the baseline model. Interestingly, the model fine-tuned with 2 months of data is less sensitive to this behaviour and detects the anomaly a week later. Nevertheless, the anomaly could be detected weeks before maintenance takes place. Also, the anomaly score shows a positive trend, before the threshold is exceeded.

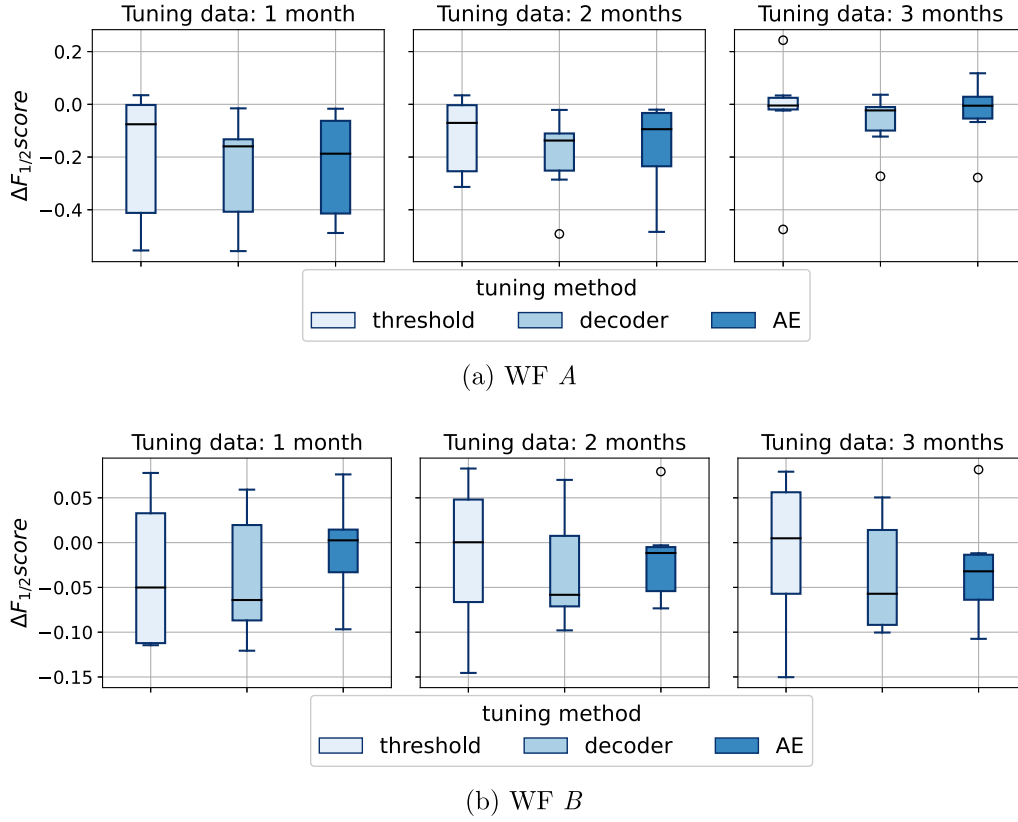


Fig. 4. Transfer learning results using data of multiple wind turbines to train a source model and fine-tuning on the target wind turbine. The difference in $F_{1/2}$ score with respect to the baseline model for each wind turbine is shown.

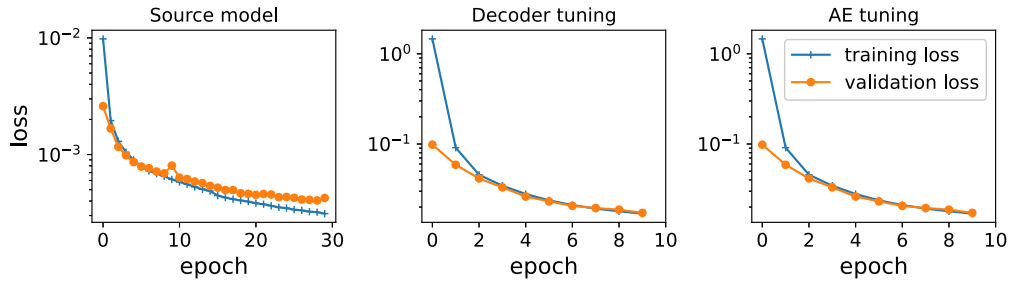


Fig. 5. Learning curves of a *multi-asset* source model and for the tuning methods decoder and AE for a wind turbine of WF A.

4.3.3. Case study 3

Anomalies are detected between 2023-02-28 and 2023-03-27 for a WT in WF B, as shown in Fig. 8. The cause of the anomalies was a failure in the gearbox of the WT. Both TL models are able to detect some anomalous behaviour on 2023-03-01, after which the anomaly score normalises for a short time. Next, the WT is regularly in derated operation. Afterwards the turbine was down for maintenance and the gearbox was replaced.

4.4. Discussion

The models used in this study are fairly straight-forward to focus on the application of transfer learning to overcome scarce training data for AE-based anomaly detection. The performance of the models can likely be improved using more advanced AE models, such as recurrent or convolutional neural networks that consider the temporal dependencies

in the data [13,15,17] or variational autoencoders, which allow for control of the latent space [17].

Additionally, NBMs need a clean training data, i.e. without anomalies, to be able to learn normal behaviour well. This cannot always be guaranteed, for example due to annotation errors, but also because annotation by domain experts is usually too expensive in large datasets. AE models can be very sensitive to anomalies in the training data, which calls for research into methods to make AE more robust to unclean training data, such as in [28].

Finally, while we have shown that anomaly detection in WTs is possible with a limited amount of training data, AE models are a type of black-box models. A key challenge to adopt such models in the wind industry is the lack of transparency [29,30]. Future research should therefore focus on developing explainable AI frameworks such as developed in [30] for high-dimensional data or applying algorithms for explaining AE output such as developed in [16,31].

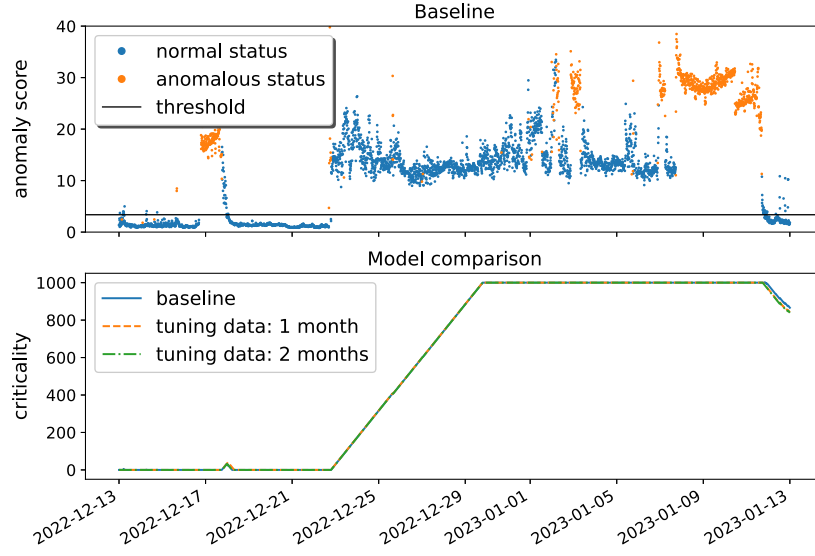


Fig. 6. Case study 1: Rotor brake and hydraulic problems for a wind turbine of WF A cause a change point in some sensor measurements. The anomalous behaviour is detected correctly by TL models using only 1 to 2 months of data for fine-tuning. The upper graph depicts the anomaly score and threshold for the baseline model, the lower graph compares the criticality for the three models.

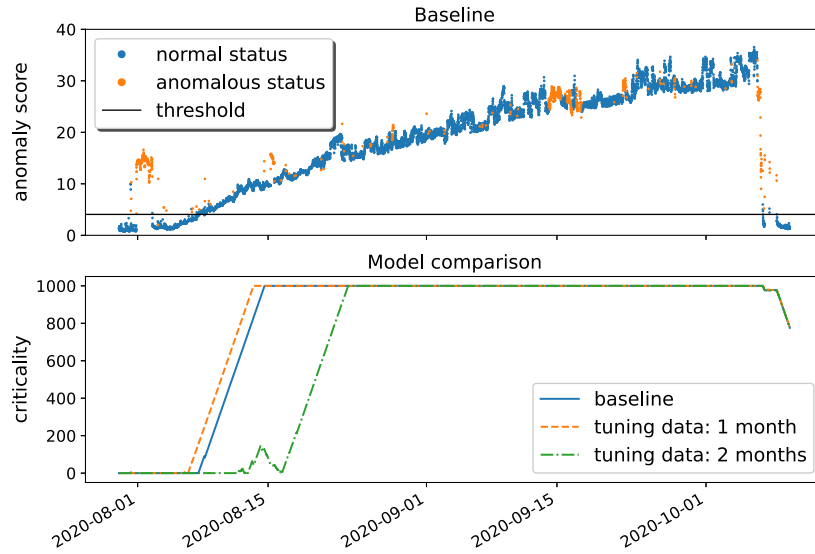


Fig. 7. Case study 2: Sensor drift caused by problems in the water cooling system of a wind turbine of WF A cause a change point in some sensor measurements. The anomalous behaviour is detected correctly by TL models using only 1 to 2 months of data for fine-tuning. The upper graph depicts the anomaly score and threshold for the baseline model, the lower graph compares the criticality for the three models.

5. Conclusions

To investigate the feasibility of AE-based anomaly detection with limited training data for WTs, two transfer learning applications using tree methods were conducted. The study focused on anomaly detection within a single wind farm, using different amounts of tuning data and WT types from two wind farms.

Two types of TL models were studied, *asset-to-asset* and *multi-asset* TL models. In case of *asset-to-asset* the source models were trained on a single WT and fine-tuned on data of the target WT. For the *multi-asset* TL models, the source models were trained on data of multiple WTs within the same wind farm and fine-tuned on data of the target WT.

In both cases, the TL models fine-tuned using 1 to 3 months of data performed as well or slightly better than the baseline models, which were trained on a full year of data. The *asset-to-asset* TL models showed superior performance compared to the *multi-asset* TL models. Additionally, three TL methods were investigated: only updating the threshold, fine-tuning the decoder and fine-tuning the AE, of which the decoder-method showed the best results.

To demonstrate that early fault detection in WTs is indeed possible even with a limited amount of training data three case studies were conducted. In all three cases the failure was detected by TL models tuned on only 1 to 2 months of data, highlighting the potential of TL for improving anomaly detection in WTs, reducing the amount of data and resources needed for effective detection.

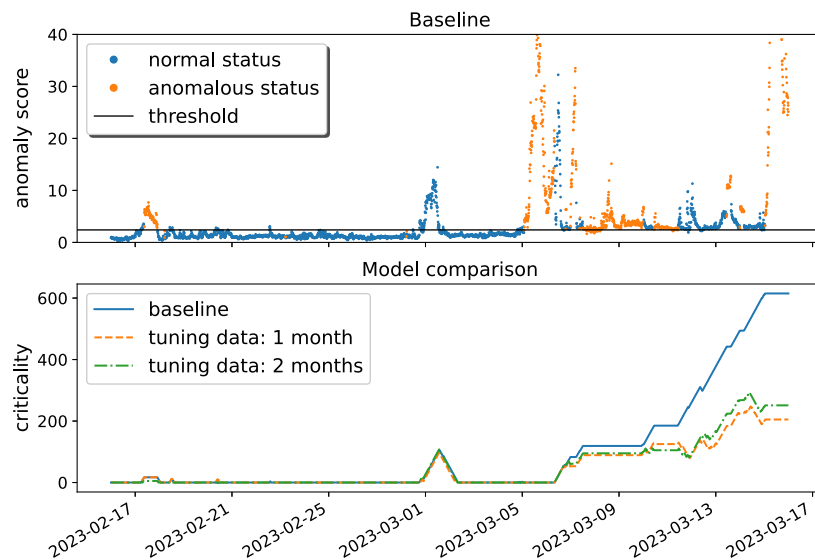


Fig. 8. Case study 3: Anomalies caused by a failure in the gearbox of a wind turbine of WF B. Using only 1 to 2 months of data for fine-tuning, the anomaly detection model is capable of detecting the failure correctly. The upper graph depicts the anomaly score and threshold for the baseline model, the lower graph compares the criticality for the three models.

CRedit authorship contribution statement

Cyriana M.A. Roelofs: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Christian Gück:** Writing – review & editing, Validation, Software. **Stefan Faulstich:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

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